Brown representability under detecting and countably factorizing generators

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Editorial note. Original theorem, complete author proof, surrounding source sections and shared bibliography retained. Cross-section/chapter references outside the collected sections display their exact original labels, without invented theorem numbers. No hypothesis or mathematical argument added or repaired.

Allowed standard prerequisites (editorial).

- Additive categories, categories of fractions, triangulated categories and basic homological algebra. All selected theorem constructions and the surrounding local definitions and proofs are included. Standard external references retain their exact original labels and shared bibliography.. No primary localization or representing-object construction is assumed as background.

1 Derived colimits

In a triangulated category there is a notion of derived colimit.

Definition 1.1. Let $\mathcal{D}$ be a triangulated category. Let (K_n, f_n) be a system of objects of $\mathcal{D}$. We say an object K is a derived colimit, or a homotopy colimit of the system (K_n) if the direct sum $\bigoplus K_n$ exists and there is a distinguished triangle

$$\bigoplus K_n \to \bigoplus K_n \to K \to \bigoplus K_n[1]$$

where the map $\bigoplus K_n \to \bigoplus K_n$ is given by $1 - f_n$ in degree n. If this is the case, then we sometimes indicate this by the notation $K = \mathrm{hocolim} K_n$.

By TR3 a derived colimit, if it exists, is unique up to (non-unique) isomorphism. Moreover, by TR1 a derived colimit of K_n exists as soon as $\bigoplus K_n$ exists. The derived category $D(\mathrm{Ab})$ of the category of abelian groups is an example of a triangulated category where all homotopy colimits exist.

The nonuniqueness makes it hard to pin down the derived colimit. In More on Algebra, Lemma [original label: `more-algebra-lemma-map-from-hocolim`] the reader finds an exact sequence

$$0 \to R^1 \lim \mathrm{Hom}(K_n, L[-1]) \to \mathrm{Hom}(\mathrm{hocolim} K_n, L) \to \lim \mathrm{Hom}(K_n, L) \to 0$$

describing the Homs out of a homotopy colimit in terms of the usual Homs.

Remark 1.2. Let $\mathcal{D}$ be a triangulated category. Let (K_n, f_n) be a system of objects of $\mathcal{D}$. We may think of a derived colimit as an object K of $\mathcal{D}$ endowed with morphisms $i_n : K_n \to K$ such that $i_{n+1} \circ f_n = i_n$ and such that there exists a morphism $c : K \to \bigoplus K_n[1]$ with the property that

$$\bigoplus K_n \xrightarrow{1-f_n} \bigoplus K_n \xrightarrow{i_n} K \xrightarrow{c} \bigoplus K_n[1]$$

is a distinguished triangle. If (K', i'_n, c') is a second derived colimit, then there exists an isomorphism $\varphi : K \to K'$ such that $\varphi \circ i_n = i'_n$ and $c' \circ \varphi = c$. The existence of φ is TR3 and the fact that φ is an isomorphism is Lemma [original label: `lemma-third-isomorphism-triangle`].

Remark 1.3. Let $\mathcal{D}$ be a triangulated category. Let $(a_n) : (K_n, f_n) \to (L_n, g_n)$ be a morphism of systems of objects of $\mathcal{D}$. Let (K, i_n, c) be a derived colimit of the first system and let (L, j_n, d) be a derived colimit of the second system with notation as in Remark 1.2. Then there exists a morphism $a : K \to L$ such that $a \circ i_n = j_n$ and $d \circ a = (a_n[1]) \circ c$. This follows from TR3 applied to the defining distinguished triangles.

Lemma 1.4. Let $\mathcal{D}$ be a triangulated category. Let (K_n, f_n) be a system of objects of $\mathcal{D}$. Let $n_1 < n_2 < n_3 < \ldots$ be a sequence of integers. Assume $\bigoplus K_n$ and $\bigoplus K_{n_i}$ exist. Then there exists an isomorphism $\mathrm{hocolim} K_{n_i} \to \mathrm{hocolim} K_n$ such that

$$\begin{array}{ccc} K_{n_i} & \longrightarrow & \mathrm{hocolim} K_{n_i} \\ {\scriptstyle \mathrm{id}}\downarrow & & \downarrow \\ K_{n_i} & \longrightarrow & \mathrm{hocolim} K_n \end{array}$$

commutes for all i.

Proof. Let $g_i : K_{n_i} \to K_{n_{i+1}}$ be the composition $f_{n_{i+1}-1} \circ \ldots \circ f_{n_i}$. We construct commutative diagrams

$$\begin{array}{ccc} \bigoplus_i K_{n_i} & \xrightarrow{1-g_i} & \bigoplus_i K_{n_i} \\ {\scriptstyle b}\downarrow & & \downarrow{\scriptstyle a} \\ \bigoplus_n K_n & \xrightarrow{1-f_n} & \bigoplus_n K_n \end{array} \quad \text{and} \quad \begin{array}{ccc} \bigoplus_n K_n & \xrightarrow{1-f_n} & \bigoplus_n K_n \\ {\scriptstyle d}\downarrow & & \downarrow{\scriptstyle c} \\ \bigoplus_i K_{n_i} & \xrightarrow{1-g_i} & \bigoplus_i K_{n_i} \end{array}$$

as follows. Let $a_i = a|_{K_{n_i}}$ be the inclusion of K_{n_i} into the direct sum. In other words, a is the natural inclusion. Let $b_i = b|_{K_{n_i}}$ be the map

$$K_{n_i} \xrightarrow{1,\ f_{n_i},\ f_{n_i+1} \circ f_{n_i},\ \ldots,\ f_{n_{i+1}-2} \circ \ldots \circ f_{n_i}} K_{n_i} \oplus K_{n_i+1} \oplus \ldots \oplus K_{n_{i+1}-1}$$

If $n_{i-1} < j \leq n_i$, then we let $c_j = c|_{K_j}$ be the map

$$K_j \xrightarrow{f_{n_i-1} \circ \ldots \circ f_j} K_{n_i}$$

We let $d_j = d|_{K_j}$ be zero if $j \neq n_i$ for any i and we let d_{n_i} be the natural inclusion of K_{n_i} into the direct sum. In other words, d is the natural projection. By TR3 these diagrams define morphisms

$$\varphi : \mathrm{hocolim} K_{n_i} \to \mathrm{hocolim} K_n \quad \text{and} \quad \psi : \mathrm{hocolim} K_n \to \mathrm{hocolim} K_{n_i}$$

Since $c \circ a$ and $d \circ b$ are the identity maps we see that $\varphi \circ \psi$ is an isomorphism by Lemma [original label: `lemma-third-isomorphism-triangle`]. The other way around we get the morphisms $a \circ c$ and $b \circ d$. Consider the morphism $h = (h_j) : \bigoplus K_n \to \bigoplus K_n$ given by the rule: for $n_{i-1} < j < n_i$ we set

$$h_j : K_j \xrightarrow{1,\ f_j,\ f_{j+1} \circ f_j,\ \ldots,\ f_{n_i - 1} \circ \ldots \circ f_j} K_j \oplus \ldots \oplus K_{n_i}$$

Then the reader verifies that $(1 - f) \circ h = \mathrm{id} - a \circ c$ and $h \circ (1 - f) = \mathrm{id} - b \circ d$. This means that $\mathrm{id} - \psi \circ \varphi$ has square zero by Lemma [original label: `lemma-third-map-square-zero`] (small argument omitted). In other words, $\psi \circ \varphi$ differs from the identity by a nilpotent endomorphism, hence is an isomorphism. Thus φ and ψ are isomorphisms as desired. □

Lemma 1.5. Let $\mathcal{A}$ be an abelian category. If $\mathcal{A}$ has exact countable direct sums, then $D(\mathcal{A})$ has countable direct sums. In fact given a collection of complexes $K_i^\bullet$ indexed by a countable index set I the termwise direct sum $\bigoplus K_i^\bullet$ is the direct sum of $K_i^\bullet$ in $D(\mathcal{A})$.

Proof. Let $L^\bullet$ be a complex. Suppose given maps $\alpha_i : K_i^\bullet \to L^\bullet$ in $D(\mathcal{A})$. This means there exist quasi-isomorphisms $s_i : M_i^\bullet \to K_i^\bullet$ of complexes and maps of complexes $f_i : M_i^\bullet \to L^\bullet$ such that $\alpha_i = f_i s_i^{-1}$. By assumption the map of complexes

$$s : \bigoplus M_i^\bullet \longrightarrow \bigoplus K_i^\bullet$$

is a quasi-isomorphism. Hence setting $f = \bigoplus f_i$ we see that $\alpha = f s^{-1}$ is a map in $D(\mathcal{A})$ whose composition with the coprojection $K_i^\bullet \to \bigoplus K_i^\bullet$ is α_i. We omit the verification that α is unique. □

Lemma 1.6. Let $\mathcal{A}$ be an abelian category. Assume colimits over N exist and are exact. Then countable direct sums exists and are exact. Moreover, if (A_n, f_n) is a system over N, then there is a short exact sequence

$$0 \to \bigoplus A_n \to \bigoplus A_n \to \operatorname{colim} A_n \to 0$$

where the first map in degree n is given by $1 - f_n$.

Proof. The first statement follows from $\bigoplus A_n = \operatorname{colim}(A_1 \oplus \ldots \oplus A_n)$. For the second, note that for each n we have the short exact sequence

$$0 \to A_1 \oplus \ldots \oplus A_{n-1} \to A_1 \oplus \ldots \oplus A_n \to A_n \to 0$$

where the first map is given by the maps $1 - f_i$ and the second map is the sum of the transition maps. Take the colimit to get the sequence of the lemma. □

Lemma 1.7. Let $\mathcal{A}$ be an abelian category. Let $L_n^\bullet$ be a system of complexes of $\mathcal{A}$. Assume colimits over N exist and are exact in $\mathcal{A}$. Then the termwise colimit $L^\bullet = \operatorname{colim} L_n^\bullet$ is a homotopy colimit of the system in $D(\mathcal{A})$.

Proof. We have an exact sequence of complexes

$$0 \to \bigoplus L_n^\bullet \to \bigoplus L_n^\bullet \to L^\bullet \to 0$$

by Lemma 1.6. The direct sums are direct sums in $D(\mathcal{A})$ by Lemma 1.5. Thus the result follows from the definition of derived colimits in Definition 1.1 and the fact that a short exact sequence of complexes gives a distinguished triangle (Lemma [original label: `lemma-derived-canonical-delta-functor`]). □

Lemma 1.8. Let $\mathcal{D}$ be a triangulated category having countable direct sums. Let $\mathcal{A}$ be an abelian category with exact colimits over N. Let $H : \mathcal{D} \to \mathcal{A}$ be a homological functor commuting with countable direct sums. Then $H(\text{hocolim} K_n) = \text{colim}\, H(K_n)$ for any system of objects of $\mathcal{D}$.

Proof. Write $K = \text{hocolim} K_n$. Apply H to the defining distinguished triangle to get

$$\bigoplus H(K_n) \to \bigoplus H(K_n) \to H(K) \to \bigoplus H(K_n[1]) \to \bigoplus H(K_n[1])$$

where the first map is given by $1 - H(f_n)$ and the last map is given by $1 - H(f_n[1])$. Apply Lemma 1.6 to see that this proves the lemma. □

The following lemma tells us that taking maps out of a compact object (to be defined later) commutes with derived colimits.

Lemma 1.9. Let $\mathcal{D}$ be a triangulated category with countable direct sums. Let $K \in \mathcal{D}$ be an object such that for every countable set of objects $E_n \in \mathcal{D}$ the canonical map

$$\bigoplus \text{Hom}_{\mathcal{D}}(K, E_n) \longrightarrow \text{Hom}_{\mathcal{D}}(K, \bigoplus E_n)$$

is a bijection. Then, given any system L_n of $\mathcal{D}$ over N whose derived colimit $L = \text{hocolim} L_n$ exists we have that

$$\text{colim}\, \text{Hom}_{\mathcal{D}}(K, L_n) \longrightarrow \text{Hom}_{\mathcal{D}}(K, L)$$

is a bijection.

Proof. By Lemma [original label: `lemma-representable-homological`], this is a particular case of Lemma 1.8. □

2 Derived limits

In a triangulated category there is a notion of derived limit.

Definition 2.1. Let $\mathcal{D}$ be a triangulated category. Let (K_n, f_n) be an inverse system of objects of $\mathcal{D}$. We say an object K is a derived limit, or a homotopy limit of the system (K_n) if the product $\prod K_n$ exists and there is a distinguished triangle

$$K \to \prod K_n \to \prod K_n \to K[1]$$

where the map $\prod K_n \to \prod K_n$ is given by $(k_n) \mapsto (k_n - f_{n+1}(k_{n+1}))$. If this is the case, then we sometimes indicate this by the notation $K = R\lim K_n$.

By TR3 a derived limit, if it exists, is unique up to (non-unique) isomorphism. Moreover, by TR1 a derived limit $R\lim K_n$ exists as soon as $\prod K_n$ exists. The derived category $D(\text{Ab})$ of the category of abelian groups is an example of a triangulated category where all derived limits exist.

The nonuniqueness makes it hard to pin down the derived limit. In More on Algebra, Lemma [original label: `more-algebra-lemma-map-into-Rlim`] the reader finds an exact sequence

$$0 \to R^1 \lim \text{Hom}(L, K_n[-1]) \to \text{Hom}(L, R\lim K_n) \to \lim \text{Hom}(L, K_n) \to 0$$

describing the Homs into a derived limit in terms of the usual Homs.

Lemma 2.2. Let $\mathcal{A}$ be an abelian category with exact countable products. Then

1. $D(\mathcal{A})$ has countable products,
2. countable products $\prod K_i$ in $D(\mathcal{A})$ are obtained by taking termwise products of any complexes representing the K_i, and
3. $H^p(\prod K_i) = \prod H^p(K_i)$.

Proof. Let $K_i^\bullet$ be a complex representing K_i in $D(\mathcal{A})$. Let $L^\bullet$ be a complex. Suppose given maps $\alpha_i : L^\bullet \to K_i^\bullet$ in $D(\mathcal{A})$. This means there exist quasi-isomorphisms $s_i : K_i^\bullet \to M_i^\bullet$ of complexes and maps of complexes $f_i : L^\bullet \to M_i^\bullet$ such that $\alpha_i = s_i^{-1} f_i$. By assumption the map of complexes

$$s : \prod K_i^\bullet \longrightarrow \prod M_i^\bullet$$

is a quasi-isomorphism. Hence setting $f = \prod f_i$ we see that $\alpha = s^{-1} f$ is a map in $D(\mathcal{A})$ whose composition with the projection $\prod K_i^\bullet \to K_i^\bullet$ is α_i. We omit the verification that α is unique. □

The duals of Lemmas 1.6, 1.7, and 1.9 should be stated here and proved. However, we do not know any applications of these lemmas for now.

Lemma 2.3. Let $\mathcal{A}$ be an abelian category with countable products and enough injectives. Let (K_n) be an inverse system of $D^+(\mathcal{A})$. Then $R\lim K_n$ exists.

Proof. It suffices to show that $\prod K_n$ exists in $D(\mathcal{A})$. For every n we can represent K_n by a bounded below complex $I_n^\bullet$ of injectives (Lemma [original label: `lemma-injective-resolutions-exist`]). Then $\prod K_n$ is represented by $\prod I_n^\bullet$, see Lemma [original label: `lemma-product-K-injective`]. □

Remark 2.4. Let $\mathcal{D}$ be a triangulated category. Let (K_n) and (L_n) be inverse systems of objects of $\mathcal{D}$ with derived limits K and L. Suppose we have a morphism $a : (K_n) \to (L_n)$ of pro-objects, see Categories, Example [original label: `categories-example-pro-morphism-inverse-systems`]. This means we are given maps $a_n : K_{m(n)} \to L_n$. We may assume $m(1) < m(2) < m(3) < \ldots$ and that the maps a_n are compatible with the transition maps in the systems. Then we can consider the maps

$$a', a'' : \prod K_n \longrightarrow \prod L_n$$

defined by the rules $a'((k_n)) = (a_n(k_{m(n)}))$ and

$$a''((k_n)) = (a_n(k_{m(n)} + f(k_{m(n)+1}) + \ldots + f(k_{m(n+1)-1})))$$

where each occurence of f denotes a suitable transition map of the inverse system (K_n). Then the solid diagram

$$\begin{array}{ccccc} K & \longrightarrow & \prod K_n & \longrightarrow & \prod K_n \\ \vdots & & \downarrow{\scriptstyle a'} & & \downarrow{\scriptstyle a''} \\ L & \longrightarrow & \prod L_n & \longrightarrow & \prod L_n \end{array}$$

is commutative and by TR3 we obtain the dotted arrow producing a morphism of distinguished triangles. We warn the reader that the map $K \to L$ is not unique. We will see later, that if a is a pro-isomorphism, then $K \to L$ is an isomorphism, see More on Algebra, Lemma [original label: `more-algebra-lemma-pro-isomorphism-Rlim`].

Remark 2.5. Let $\mathcal{A}$ be an abelian category. Let $K^\bullet$ be a complex of $\mathcal{A}$. Then $\tau_{\geq -n}K^\bullet$ is an inverse system of complexes and which in particular determines an inverse system in $D(\mathcal{A})$. Let us assume that $R\lim \tau_{\geq -n}K^\bullet$ exists. Then the canonical maps $c_n : K^\bullet \to \tau_{\geq -n}K^\bullet$ are compatible with the transition maps of our inverse system. By the defining distinguished triangle of Definition 2.1 and Lemma [original label: `lemma-representable-homological`] we conclude there exists a morphism

$$c : K^\bullet \longrightarrow R\lim \tau_{\geq -n}K^\bullet$$

in $D(\mathcal{A})$ such that the composition of c with the projection $R\lim \tau_{\geq -n}K^\bullet \to \tau_{\geq -m}K^\bullet$ is equal to c_m. Now the morphism c may not be unique, but we claim that whether or not c is an isomorphism is independent of the choice of c (and of our choice of the homotopy limit). Namely, for $i \in \mathbf{Z}$ and for $m > -i$ the composition

$$H^i(K^\bullet) \xrightarrow{H^i(c)} H^i(R\lim \tau_{\geq -n}K^\bullet) \to H^i(\tau_{\geq -m}K^\bullet) = H^i(K^\bullet)$$

is the identity. Hence $H^i(c)$ is an isomorphism if and only if the second map is an isomorphism. This is independent of c and also independent of the choice of the homotopy limit (as any two choices are isomorphic).

Lemma 2.6. Let $\mathcal{A}$ be an abelian category with countable products and enough injectives. Let $K^\bullet$ be a complex. Let $I_n^\bullet$ be the inverse system of bounded below complexes of injectives produced by Lemma [original label: `lemma-special-inverse-system`]. Then $I^\bullet = \lim I_n^\bullet$ exists, is K-injective, represents $R\lim \tau_{\geq -n}K^\bullet$ in $D(\mathcal{A})$, and the following are equivalent

1. the map $K^\bullet \to I^\bullet$ (see proof) is a quasi-isomorphism,
2. the map $K^\bullet \to R\lim \tau_{\geq -n}K^\bullet$ of Remark 2.5 is an isomorphism in $D(\mathcal{A})$.

Proof. The statement of the lemma makes sense as $R\lim \tau_{\geq -n}K^\bullet$ exists by Lemma 2.3. Each complex $I_n^\bullet$ is K-injective by Lemma [original label: `lemma-bounded-below-injectives-K-`
Choose direct sum decompositions $I_{n+1}^p = C_{n+1}^p \oplus I_n^p$ for all $n \geq 1$. Set $C_1^p = I_1^p$. The complex $I^\bullet = \lim I_n^\bullet$ exists because we can take $I^p = \prod_{n \geq 1} C_n^p$. Fix $p \in \mathbf{Z}$. We claim there is a split short exact sequence

$$0 \to I^p \to \prod I_n^p \to \prod I_n^p \to 0$$

of objects of $\mathcal{A}$. Here the first map is given by the projection maps $I^p \to I_n^p$ and the second map by $(x_n) \mapsto (x_n - f_{n+1}^p(x_{n+1}))$ where $f_n^p : I_n^p \to I_{n-1}^p$ are the transition maps. The splitting comes from the map $\prod I_n^p \to \prod C_n^p = I^p$. We obtain a termwise split short exact sequence of complexes

$$0 \to I^\bullet \to \prod I_n^\bullet \to \prod I_n^\bullet \to 0$$

Hence a corresponding distinguished triangle in $K(\mathcal{A})$ and $D(\mathcal{A})$. By Lemma [original label: `lemma-product-K-injective`] the products are K-injective and

represent the corresponding products in $D(\mathcal{A})$. It follows that $I^\bullet$ represents $R\lim I_n^\bullet$ (Definition 2.1). Since $R\lim I_n^\bullet \cong R\lim \tau_{\geq -n}K^\bullet$ as derived limits are defined on the level of the derived category, we see that $I^\bullet$ represents $R\lim \tau_{\geq -n}K^\bullet$. Moreover, the complex $I^\bullet$ is K-injective by Lemma [original label: `lemma-triangle-K-injective`]. By the commutative diagram of Lemma [original label: `lemma-special-inverse-system`] and since $K^i = (\tau_{\geq -n}K^\bullet)^i$ for $n \gg 0$ we see that we get a unique map $\gamma : K^\bullet \to I^\bullet$ such that the diagrams

$$\begin{array}{ccc} K^\bullet & \longrightarrow & \tau_{\geq -n}K^\bullet \\ {\scriptstyle\gamma}\downarrow & & \downarrow \\ I^\bullet & \longrightarrow & I_n^\bullet \end{array}$$

commute. It follows that γ is a map of complexes which represents the map $c : K^\bullet \to R\lim \tau_{\geq -n}K^\bullet$ of Remark 2.5 in $D(\mathcal{A})$. In other words, the diagram

$$\begin{array}{ccc} K^\bullet & \xrightarrow[c]{} & R\lim \tau_{\geq -n}K^\bullet \\ {\scriptstyle\gamma}\downarrow & & \downarrow{\scriptstyle\cong} \\ I^\bullet & \xrightarrow{\cong} & R\lim I_n^\bullet \end{array}$$

is commutative in $D(\mathcal{A})$. The lemma follows. □

Lemma 2.7. Let $\mathcal{A}$ be an abelian category having enough injectives and exact countable products. Then for every complex there is a quasi-isomorphism to a K-injective complex.

Proof. By Lemma 2.6 it suffices to show that $K \to R\lim \tau_{\geq -n}K$ is an isomorphism for all K in $D(\mathcal{A})$. Consider the defining distinguished triangle

$$R\lim \tau_{\geq -n}K \to \prod \tau_{\geq -n}K \to \prod \tau_{\geq -n}K \to (R\lim \tau_{\geq -n}K)[1]$$

By Lemma 2.2 we have

$$H^p(\prod \tau_{\geq -n}K) = \prod\nolimits_{p \geq -n} H^p(K)$$

It follows in a straightforward manner from the long exact cohomology sequence of the displayed distinguished triangle that $H^p(R\lim \tau_{\geq -n}K) = H^p(K)$. □

3 Operations on full subcategories

Let $\mathcal{T}$ be a triangulated category. We will identify full subcategories of $\mathcal{T}$ with subsets of $\mathrm{Ob}(\mathcal{T})$. Given full subcategories $\mathcal{A}, \mathcal{B}, \ldots$ we let

1. $\mathcal{A}[a, b]$ for $-\infty \leq a \leq b \leq \infty$ be the full subcategory of $\mathcal{T}$ consisting of all objects $A[-i]$ with $i \in [a, b] \cap \mathrm{Z}$ with $A \in \mathrm{Ob}(\mathcal{A})$ (note the minus sign!),
2. $smd(\mathcal{A})$ be the full subcategory of $\mathcal{T}$ consisting of all objects which are isomorphic to direct summands of objects of $\mathcal{A}$,
3. $add(\mathcal{A})$ be the full subcategory of $\mathcal{T}$ consisting of all objects which are isomorphic to finite direct sums of objects of $\mathcal{A}$,

4. $\mathcal{A} \star \mathcal{B}$ be the full subcategory of $\mathcal{T}$ consisting of all objects X of $\mathcal{T}$ which fit into a distinguished triangle $A \to X \to B$ with $A \in \mathrm{Ob}(\mathcal{A})$ and $B \in \mathrm{Ob}(\mathcal{B})$,

5. $\mathcal{A}^{\star n} = \mathcal{A} \star \ldots \star \mathcal{A}$ with $n \geq 1$ factors (we will see $\star$ is associative below),

6. $smd(add(\mathcal{A})^{\star n}) = smd(add(\mathcal{A}) \star \ldots \star add(\mathcal{A}))$ with $n \geq 1$ factors.

If E is an object of $\mathcal{T}$, then we think of E sometimes also as the full subcategory of $\mathcal{T}$ whose single object is E. Then we can consider things like $add(E[-1, 2])$ and so on and so forth. We warn the reader that this notation is not universally accepted.

Lemma 3.1. Let $\mathcal{T}$ be a triangulated category. Given full subcategories $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ we have $(\mathcal{A} \star \mathcal{B}) \star \mathcal{C} = \mathcal{A} \star (\mathcal{B} \star \mathcal{C})$.

Proof. If we have distinguished triangles $A \to X \to B$ and $X \to Y \to C$ then by Axiom TR4 we have distinguished triangles $A \to Y \to Z$ and $B \to Z \to C$. □

Lemma 3.2. Let $\mathcal{T}$ be a triangulated category. Given full subcategories $\mathcal{A}$, $\mathcal{B}$ we have $smd(\mathcal{A}) \star smd(\mathcal{B}) \subset smd(\mathcal{A} \star \mathcal{B})$ and $smd(smd(\mathcal{A}) \star smd(\mathcal{B})) = smd(\mathcal{A} \star \mathcal{B})$.

Proof. Suppose we have a distinguished triangle $A_1 \to X \to B_1$ where $A_1 \oplus A_2 \in \mathrm{Ob}(\mathcal{A})$ and $B_1 \oplus B_2 \in \mathrm{Ob}(\mathcal{B})$. Then we obtain a distinguished triangle $A_1 \oplus A_2 \to A_2 \oplus X \oplus B_2 \to B_1 \oplus B_2$ which proves that X is in $smd(\mathcal{A} \star \mathcal{B})$. This proves the inclusion. The equality follows trivially from this. □

Lemma 3.3. Let $\mathcal{T}$ be a triangulated category. Given full subcategories $\mathcal{A}$, $\mathcal{B}$ the full subcategories $add(\mathcal{A}) \star add(\mathcal{B})$ and $smd(add(\mathcal{A}))$ are closed under direct sums.

Proof. Namely, if $A \to X \to B$ and $A' \to X' \to B'$ are distinguished triangles and $A, A' \in add(\mathcal{A})$ and $B, B' \in add(\mathcal{B})$ then $A \oplus A' \to X \oplus X' \to B \oplus B'$ is a distinguished triangle with $A \oplus A' \in add(\mathcal{A})$ and $B \oplus B' \in add(\mathcal{B})$. The result for $smd(add(\mathcal{A}))$ is trivial. □

Lemma 3.4. Let $\mathcal{T}$ be a triangulated category. Given a full subcategory $\mathcal{A}$ for $n \geq 1$ the subcategory

$$\mathcal{C}_n = smd(add(\mathcal{A})^{\star n}) = smd(add(\mathcal{A}) \star \ldots \star add(\mathcal{A}))$$

defined above is a strictly full subcategory of $\mathcal{T}$ closed under direct sums and direct summands and $\mathcal{C}_{n+m} = smd(\mathcal{C}_n \star \mathcal{C}_m)$ for all $n, m \geq 1$.

Proof. Immediate from Lemmas 3.1, 3.2, and 3.3. □

Remark 3.5. Let $F : \mathcal{T} \to \mathcal{T}'$ be an exact functor of triangulated categories. Given a full subcategory $\mathcal{A}$ of $\mathcal{T}$ we denote $F(\mathcal{A})$ the full subcategory of $\mathcal{T}'$ whose objects consists of all objects $F(A)$ with $A \in \mathrm{Ob}(\mathcal{A})$. We have

$$F(\mathcal{A}[a, b]) = F(\mathcal{A})[a, b]$$

$$F(smd(\mathcal{A})) \subset smd(F(\mathcal{A})),$$

$$F(add(\mathcal{A})) \subset add(F(\mathcal{A})),$$

$$F(\mathcal{A} \star \mathcal{B}) \subset F(\mathcal{A}) \star F(\mathcal{B}),$$
$$F(\mathcal{A}^{\star n}) \subset F(\mathcal{A})^{\star n}.$$

We omit the trivial verifications.

Remark 3.6. Let $\mathcal{T}$ be a triangulated category. Given full subcategories $\mathcal{A}_1 \subset \mathcal{A}_2 \subset \mathcal{A}_3 \subset \ldots$ and $\mathcal{B}$ of $\mathcal{T}$ we have

$$\left(\bigcup \mathcal{A}_i\right)[a,b] = \bigcup \mathcal{A}_i[a,b]$$
$$smd\left(\bigcup \mathcal{A}_i\right) = \bigcup smd(\mathcal{A}_i),$$
$$add\left(\bigcup \mathcal{A}_i\right) = \bigcup add(\mathcal{A}_i),$$
$$\left(\bigcup \mathcal{A}_i\right) \star \mathcal{B} = \bigcup \mathcal{A}_i \star \mathcal{B},$$
$$\mathcal{B} \star \left(\bigcup \mathcal{A}_i\right) = \bigcup \mathcal{B} \star \mathcal{A}_i,$$
$$\left(\bigcup \mathcal{A}_i\right)^{\star n} = \bigcup \mathcal{A}_i^{\star n}.$$

We omit the trivial verifications.

Lemma 3.7. Let $\mathcal{A}$ be an abelian category. Let $\mathcal{D} = D(\mathcal{A})$. Let $\mathcal{E} \subset \mathrm{Ob}(\mathcal{A})$ be a subset which we view as a subset of $\mathrm{Ob}(\mathcal{D})$ also. Let K be an object of $\mathcal{D}$.

1. Let $b \geq a$ and assume $H^i(K)$ is zero for $i \notin [a,b]$ and $H^i(K) \in \mathcal{E}$ if $i \in [a,b]$. Then K is in $smd(add(\mathcal{E}[a,b])^{\star(b-a+1)})$.
2. Let $b \geq a$ and assume $H^i(K)$ is zero for $i \notin [a,b]$ and $H^i(K) \in smd(add(\mathcal{E}))$ if $i \in [a,b]$. Then K is in $smd(add(\mathcal{E}[a,b])^{\star(b-a+1)})$.
3. Let $b \geq a$ and assume K can be represented by a complex $K^\bullet$ with $K^i = 0$ for $i \notin [a,b]$ and $K^i \in \mathcal{E}$ for $i \in [a,b]$. Then K is in $smd(add(\mathcal{E}[a,b])^{\star(b-a+1)})$.
4. Let $b \geq a$ and assume K can be represented by a complex $K^\bullet$ with $K^i = 0$ for $i \notin [a,b]$ and $K^i \in smd(add(\mathcal{E}))$ for $i \in [a,b]$. Then K is in $smd(add(\mathcal{E}[a,b])^{\star(b-a+1)})$.

Proof. We will use Lemma 3.4 without further mention. We will prove (2) which trivially implies (1). We use induction on $b-a$. If $b-a=0$, then K is isomorphic to $H^i(K)[-a]$ in $\mathcal{D}$ and the result is immediate. If $b-a>0$, then we consider the distinguished triangle

$$\tau_{\leq b-1}K^\bullet \to K^\bullet \to K^b[-b]$$

and we conclude by induction on $b-a$. We omit the proof of (3) and (4). □

Lemma 3.8. Let $\mathcal{T}$ be a triangulated category. Let $H : \mathcal{T} \to \mathcal{A}$ be a homological functor to an abelian category $\mathcal{A}$. Let $a \leq b$ and $\mathcal{E} \subset \mathrm{Ob}(\mathcal{T})$ be a subset such that $H^i(E) = 0$ for $E \in \mathcal{E}$ and $i \notin [a,b]$. Then for $X \in smd(add(\mathcal{E}[-m,m])^{\star n})$ we have $H^i(X) = 0$ for $i \notin [-m+na, m+nb]$.

Proof. Omitted. Pleasant exercise in the definitions. □

4 Generators of triangulated categories

In this section we briefly introduce a few of the different notions of a generator for a triangulated category. Our terminology is taken from [1] (except that we use "saturated" for what they call "épaisse", see Definition [original label: `definition-saturated`], and our definition of $add(\mathcal{A})$ is different).

Let $\mathcal{D}$ be a triangulated category. Let E be an object of $\mathcal{D}$. Denote $\langle E\rangle_1$ the strictly full subcategory of $\mathcal{D}$ consisting of objects in $\mathcal{D}$ isomorphic to direct summands of finite direct sums

$$\bigoplus\nolimits_{i=1,\ldots,r} E[n_i]$$

of shifts of E. It is clear that in the notation of Section 3 we have

$$\langle E\rangle_1 = smd(add(E[-\infty,\infty]))$$

For $n > 1$ let $\langle E\rangle_n$ denote the full subcategory of $\mathcal{D}$ consisting of objects of $\mathcal{D}$ isomorphic to direct summands of objects X which fit into a distinguished triangle

$$A \to X \to B \to A[1]$$

where A is an object of $\langle E\rangle_1$ and B an object of $\langle E\rangle_{n-1}$. In the notation of Section 3 we have

$$\langle E\rangle_n = smd(\langle E\rangle_1 \star \langle E\rangle_{n-1})$$

Each of the categories $\langle E\rangle_n$ is a strictly full additive (by Lemma 3.3) subcategory of $\mathcal{D}$ preserved under shifts and under taking summands. But, $\langle E\rangle_n$ is not necessarily closed under "taking cones" or "extensions", hence not necessarily a triangulated subcategory. This will be true for the subcategory

$$\langle E\rangle = \bigcup\nolimits_n \langle E\rangle_n$$

as will be shown in the lemmas below.

Lemma 4.1. Let $\mathcal{T}$ be a triangulated category. Let E be an object of $\mathcal{T}$. For $n \geq 1$ we have

$$\langle E\rangle_n = smd(\langle E\rangle_1 \star \ldots \star \langle E\rangle_1) = smd({\langle E\rangle_1}^{\star n}) = \bigcup\nolimits_{m\geq 1} smd(add(E[-m,m])^{\star n})$$

For $n, n' \geq 1$ we have $\langle E\rangle_{n+n'} = smd(\langle E\rangle_n \star \langle E\rangle_{n'})$.

Proof. The left equality in the displayed formula follows from Lemmas 3.1 and 3.2 and induction. The middle equality is a matter of notation. Since $\langle E\rangle_1 = smd(add(E[-\infty,\infty])])$ and since $E[-\infty,\infty] = \bigcup_{m\geq 1} E[-m,m]$ we see from Remark 3.6 and Lemma 3.2 that we get the equality on the right. Then the final statement follows from the remark and the corresponding statement of Lemma 3.4. □

Lemma 4.2. Let $\mathcal{D}$ be a triangulated category. Let E be an object of $\mathcal{D}$. The subcategory

$$\langle E\rangle = \bigcup\nolimits_n \langle E\rangle_n = \bigcup\nolimits_{n,m\geq 1} smd(add(E[-m,m])^{\star n})$$

is a strictly full, saturated, triangulated subcategory of $\mathcal{D}$ and it is the smallest such subcategory of $\mathcal{D}$ containing the object E.

Proof. The equality on the right follows from Lemma 4.1. It is clear that $\langle E \rangle = \bigcup \langle E \rangle_n$ contains E, is preserved under shifts, direct sums, direct summands. If $A \in \langle E \rangle_a$ and $B \in \langle E \rangle_b$ and if $A \to X \to B \to A[1]$ is a distinguished triangle, then $X \in \langle E \rangle_{a+b}$ by Lemma 4.1. Hence $\bigcup \langle E \rangle_n$ is also preserved under extensions and it follows that it is a triangulated subcategory.

Finally, let $\mathcal{D}' \subset \mathcal{D}$ be a strictly full, saturated, triangulated subcategory of $\mathcal{D}$ containing E. Then $\mathcal{D}'[-\infty, \infty] \subset \mathcal{D}'$, $add(\mathcal{D}) \subset \mathcal{D}'$, $smd(\mathcal{D}') \subset \mathcal{D}'$, and $\mathcal{D}' \star \mathcal{D}' \subset \mathcal{D}'$. In other words, all the operations we used to construct $\langle E \rangle$ out of E preserve $\mathcal{D}'$. Hence $\langle E \rangle \subset \mathcal{D}'$ and this finishes the proof. □

Definition 4.3. Let $\mathcal{D}$ be a triangulated category. Let E be an object of $\mathcal{D}$.

1. We say E is a classical generator of $\mathcal{D}$ if the smallest strictly full, saturated, triangulated subcategory of $\mathcal{D}$ containing E is equal to $\mathcal{D}$, in other words, if $\langle E \rangle = \mathcal{D}$.
2. We say E is a strong generator of $\mathcal{D}$ if $\langle E \rangle_n = \mathcal{D}$ for some $n \geq 1$.
3. We say E is a weak generator or a generator of $\mathcal{D}$ if for any nonzero object K of $\mathcal{D}$ there exists an integer n and a nonzero map $E \to K[n]$.

This definition can be generalized to the case of a family of objects.

Lemma 4.4. Let $\mathcal{D}$ be a triangulated category. Let E, K be objects of $\mathcal{D}$. The following are equivalent

1. $\mathrm{Hom}(E, K[i]) = 0$ for all $i \in \mathrm{Z}$,
2. $\mathrm{Hom}(E', K) = 0$ for all $E' \in \langle E \rangle$.

Proof. The implication (2) ⇒ (1) is immediate. Conversely, assume (1). Then $\mathrm{Hom}(X, K) = 0$ for all X in $\langle E \rangle_1$. Arguing by induction on n and using Lemma [original label: `lemma-representable-homological`] we see that $\mathrm{Hom}(X, K) = 0$ for all X in $\langle E \rangle_n$. □

Lemma 4.5. Let $\mathcal{D}$ be a triangulated category. Let E be an object of $\mathcal{D}$. If E is a classical generator of $\mathcal{D}$, then E is a generator.

Proof. Assume E is a classical generator. Let K be an object of $\mathcal{D}$ such that $\mathrm{Hom}(E, K[i]) = 0$ for all $i \in \mathrm{Z}$. By Lemma 4.4 $\mathrm{Hom}(E', K) = 0$ for all E' in $\langle E \rangle$. However, since $\mathcal{D} = \langle E \rangle$ we conclude that $\mathrm{id}_K = 0$, i.e., $K = 0$. □

Lemma 4.6. Let $\mathcal{D}$ be a triangulated category which has a strong generator. Let E be an object of $\mathcal{D}$. If E is a classical generator of $\mathcal{D}$, then E is a strong generator.

Proof. Let E' be an object of $\mathcal{D}$ such that $\mathcal{D} = \langle E' \rangle_n$. Since $\mathcal{D} = \langle E \rangle$ we see that $E' \in \langle E \rangle_m$ for some $m \geq 1$ by Lemma 4.2. Then $\langle E' \rangle_1 \subset \langle E \rangle_m$ hence

$$\mathcal{D} = \langle E' \rangle_n = smd(\langle E' \rangle_1 \star \ldots \star \langle E' \rangle_1) \subset smd(\langle E \rangle_m \star \ldots \star \langle E \rangle_m) = \langle E \rangle_{nm}$$

as desired. Here we used Lemma 4.1. □

Remark 4.7. Let $\mathcal{D}$ be a triangulated category. Let E be an object of $\mathcal{D}$. Let T be a property of objects of $\mathcal{D}$. Suppose that

1. if $K_i \in \mathcal{D}$, $i = 1, \ldots, r$ with $T(K_i)$ for $i = 1, \ldots, r$, then $T(\bigoplus K_i)$,
2. if $K \to L \to M \to K[1]$ is a distinguished triangle and T holds for two, then T holds for the third object,
3. if $T(K \oplus L)$ then $T(K)$ and $T(L)$, and
4. $T(E[n])$ holds for all n.

Then T holds for all objects of $\langle E \rangle$.

5 Compact objects

Here is the definition.

Definition 5.1. Let $\mathcal{D}$ be an additive category with arbitrary direct sums. A compact object of $\mathcal{D}$ is an object K such that the map

$$\bigoplus\nolimits_{i \in I} \mathrm{Hom}_{\mathcal{D}}(K, E_i) \longrightarrow \mathrm{Hom}_{\mathcal{D}}(K, \bigoplus\nolimits_{i \in I} E_i)$$

is bijective for any set I and objects $E_i \in \mathrm{Ob}(\mathcal{D})$ parametrized by $i \in I$.

This notion turns out to be very useful in algebraic geometry. It is an intrinsic condition on objects that forces the objects to be, well, compact.

Lemma 5.2. Let $\mathcal{D}$ be a (pre-)triangulated category with direct sums. Then the compact objects of $\mathcal{D}$ form the objects of a Karoubian, saturated, strictly full, (pre-)triangulated subcategory $\mathcal{D}_c$ of $\mathcal{D}$.

Proof. Let (X, Y, Z, f, g, h) be a distinguished triangle of $\mathcal{D}$ with X and Y compact. Then it follows from Lemma [original label: `lemma-representable-homological`] and the five lemma (Homology, Lemma [original label: `homology-lemma-five-lemma`]) that Z is a compact object too. It is clear that if $X \oplus Y$ is compact, then X, Y are compact objects too. Hence $\mathcal{D}_c$ is a saturated triangulated subcategory. Since $\mathcal{D}$ is Karoubian by Lemma [original label: `lemma-projectors-have-images-triangulated`] we conclude that the same is true for $\mathcal{D}_c$. □

Lemma 5.3. Let $\mathcal{D}$ be a triangulated category with direct sums. Let E_i, $i \in I$ be a family of compact objects of $\mathcal{D}$ such that $\bigoplus E_i$ generates $\mathcal{D}$. Then every object X of $\mathcal{D}$ can be written as

$$X = \mathrm{hocolim} X_n$$

where X_1 is a direct sum of shifts of the E_i and each transition morphism fits into a distinguished triangle $Y_n \to X_n \to X_{n+1} \to Y_n[1]$ where Y_n is a direct sum of shifts of the E_i.

Proof. Set $X_1 = \bigoplus_{(i,m,\varphi)} E_i[m]$ where the direct sum is over all triples (i, m, φ) such that $i \in I$, $m \in \mathrm{Z}$ and $\varphi : E_i[m] \to X$. Then X_1 comes equipped with a canonical morphism $X_1 \to X$. Given $X_n \to X$ we set $Y_n = \bigoplus_{(i,m,\varphi)} E_i[m]$ where the direct sum is over all triples (i, m, φ) such that $i \in I$, $m \in \mathrm{Z}$, and $\varphi : E_i[m] \to X_n$ is a morphism such that $E_i[m] \to X_n \to X$ is zero. Choose a distinguished triangle $Y_n \to X_n \to X_{n+1} \to Y_n[1]$ and let $X_{n+1} \to X$ be

any morphism such that $X_n \to X_{n+1} \to X$ is the given one; such a morphism exists by our choice of Y_n. We obtain a morphism $\mathrm{hocolim} X_n \to X$ by the construction of our maps $X_n \to X$. Choose a distinguished triangle

$$C \to \mathrm{hocolim} X_n \to X \to C[1]$$

Let $E_i[m] \to C$ be a morphism. Since E_i is compact, Lemma 1.9 implies the composition $E_i[m] \to C \to \mathrm{hocolim} X_n$ factors through X_n for some n, say by $E_i[m] \to X_n$. Then the construction of Y_n shows that the composition $E_i[m] \to X_n \to X_{n+1}$ is zero. In other words, the composition $E_i[m] \to C \to \mathrm{hocolim} X_n$ is zero. This means that our morphism $E_i[m] \to C$ comes from a morphism $E_i[m] \to X[-1]$. The construction of X_1 then shows that such morphism lifts to $\mathrm{hocolim} X_n$ and we conclude that our morphism $E_i[m] \to C$ is zero. The assumption that $\bigoplus E_i$ generates $\mathcal{D}$ implies that C is zero and the proof is done. □

Lemma 5.4. With assumptions and notation as in Lemma 5.3. If C is a compact object and $C \to X_n$ is a morphism, then there is a factorization $C \to E \to X_n$ where E is an object of $\langle E_{i_1} \oplus \ldots \oplus E_{i_t} \rangle$ for some $i_1, \ldots, i_t \in I$.

Proof. We prove this by induction on n. The base case $n = 1$ is clear. If $n > 1$ consider the composition $C \to X_n \to Y_{n-1}[1]$. This can be factored through some $E'[1] \to Y_{n-1}[1]$ where E' is a finite direct sum of shifts of the E_i. Let $I' \subset I$ be the finite set of indices that occur in this direct sum. Thus we obtain

$$\begin{array}{ccccccc} E' & \longrightarrow & C' & \longrightarrow & C & \longrightarrow & E'[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ Y_{n-1} & \longrightarrow & X_{n-1} & \longrightarrow & X_n & \longrightarrow & Y_{n-1}[1] \end{array}$$

By induction the morphism $C' \to X_{n-1}$ factors through $E'' \to X_{n-1}$ with E'' an object of $\langle \bigoplus_{i \in I''} E_i \rangle$ for some finite subset $I'' \subset I$. Choose a distinguished triangle

$$E' \to E'' \to E \to E'[1]$$

then E is an object of $\langle \bigoplus_{i \in I' \cup I''} E_i \rangle$. By construction and the axioms of a triangulated category we can choose morphisms $C \to E$ and a morphism $E \to X_n$ fitting into morphisms of triangles $(E', C', C) \to (E', E'', E)$ and $(E', E'', E) \to (Y_{n-1}, X_{n-1}, X_n)$. The composition $C \to E \to X_n$ may not equal the given morphism $C \to X_n$, but the compositions into Y_{n-1} are equal. Let $C \to X_{n-1}$ be a morphism that lifts the difference. By induction assumption we can factor this through a morphism $E''' \to X_{n-1}$ with E''' an object of $\langle \bigoplus_{i \in I'''} E_i \rangle$ for some finite subset $I' \subset I$. Thus we see that we get a solution on considering $E \oplus E''' \to X_n$ because $E \oplus E'''$ is an object of $\langle \bigoplus_{i \in I' \cup I'' \cup I'''} E_i \rangle$. □

Definition 5.5. Let $\mathcal{D}$ be a triangulated category with arbitrary direct sums. We say $\mathcal{D}$ is compactly generated if there exists a set E_i, $i \in I$ of compact objects such that $\bigoplus E_i$ generates $\mathcal{D}$.

The following proposition clarifies the relationship between classical generators and weak generators.

Proposition 5.6. Let $\mathcal{D}$ be a triangulated category with direct sums. Let E be a compact object of $\mathcal{D}$. The following are equivalent

1. E is a classical generator for $\mathcal{D}_c$ and $\mathcal{D}$ is compactly generated, and
2. E is a generator for $\mathcal{D}$.

Proof. If E is a classical generator for $\mathcal{D}_c$, then $\mathcal{D}_c = \langle E \rangle$. It follows formally from the assumption that $\mathcal{D}$ is compactly generated and Lemma 4.4 that E is a generator for $\mathcal{D}$.

The converse is more interesting. Assume that E is a generator for $\mathcal{D}$. Let X be a compact object of $\mathcal{D}$. Apply Lemma 5.3 with $I = \{1\}$ and $E_1 = E$ to write

$$X = \text{hocolim} X_n$$

as in the lemma. Since X is compact we find that $X \to \text{hocolim} X_n$ factors through X_n for some n (Lemma 1.9). Thus X is a direct summand of X_n. By Lemma 5.4 we see that X is an object of $\langle E \rangle$ and the lemma is proven. □

6 Brown representability

A reference for the material in this section is [4].

Lemma 6.1. Let $\mathcal{D}$ be a triangulated category with direct sums which is compactly generated. Let $H : \mathcal{D} \to \text{Ab}$ be a contravariant cohomological functor which transforms direct sums into products. Then H is representable.

Proof. Let E_i, $i \in I$ be a set of compact objects such that $\bigoplus_{i \in I} E_i$ generates $\mathcal{D}$. We may and do assume that the set of objects $\{E_i\}$ is preserved under shifts. Consider pairs (i, a) where $i \in I$ and $a \in H(E_i)$ and set

$$X_1 = \bigoplus\nolimits_{(i,a)} E_i$$

Since $H(X_1) = \prod_{(i,a)} H(E_i)$ we see that $(a)_{(i,a)}$ defines an element $a_1 \in H(X_1)$. Set $H_1 = \text{Hom}_{\mathcal{D}}(-, X_1)$. By Yoneda's lemma (Categories, Lemma [original label: `categories-lemma-yoneda`]) the element a_1 defines a natural transformation $H_1 \to H$.

We are going to inductively construct X_n and transformations $a_n : H_n \to H$ where $H_n = \text{Hom}_{\mathcal{D}}(-, X_n)$. Namely, we apply the procedure above to the functor $\text{Ker}(H_n \to H)$ to get an object

$$K_{n+1} = \bigoplus\nolimits_{(i,k),\ k \in \text{Ker}(H_n(E_i) \to H(E_i))} E_i$$

and a transformation $\text{Hom}_{\mathcal{D}}(-, K_{n+1}) \to \text{Ker}(H_n \to H)$. By Yoneda's lemma the composition $\text{Hom}_{\mathcal{D}}(-, K_{n+1}) \to H_n$ gives a morphism $K_{n+1} \to X_n$. We choose a distinguished triangle

$$K_{n+1} \to X_n \to X_{n+1} \to K_{n+1}[1]$$

in $\mathcal{D}$. The element $a_n \in H(X_n)$ maps to zero in $H(K_{n+1})$ by construction. Since H is cohomological we can lift it to an element $a_{n+1} \in H(X_{n+1})$.

We claim that $X = \text{hocolim} X_n$ represents H. Applying H to the defining distinguished triangle

$$\bigoplus X_n \to \bigoplus X_n \to X \to \bigoplus X_n[1]$$

we obtain an exact sequence

$$\prod H(X_n) \leftarrow \prod H(X_n) \leftarrow H(X)$$

Thus there exists an element $a \in H(X)$ mapping to (a_n) in $\prod H(X_n)$. Hence a natural transformation $\text{Hom}_{\mathcal{D}}(-, X) \to H$ such that

$$\text{Hom}_{\mathcal{D}}(-, X_1) \to \text{Hom}_{\mathcal{D}}(-, X_2) \to \text{Hom}_{\mathcal{D}}(-, X_3) \to \ldots \to \text{Hom}_{\mathcal{D}}(-, X) \to H$$

commutes. For each i the map $\text{Hom}_{\mathcal{D}}(E_i, X) \to H(E_i)$ is surjective, by construction of X_1. On the other hand, by construction of $X_n \to X_{n+1}$ the kernel of $\text{Hom}_{\mathcal{D}}(E_i, X_n) \to H(E_i)$ is killed by the map $\text{Hom}_{\mathcal{D}}(E_i, X_n) \to \text{Hom}_{\mathcal{D}}(E_i, X_{n+1})$. Since

$$\text{Hom}_{\mathcal{D}}(E_i, X) = \text{colim}\, \text{Hom}_{\mathcal{D}}(E_i, X_n)$$

by Lemma 1.9 we see that $\text{Hom}_{\mathcal{D}}(E_i, X) \to H(E_i)$ is injective.

To finish the proof, consider the subcategory

$$\mathcal{D}' = \{Y \in \text{Ob}(\mathcal{D}) \mid \text{Hom}_{\mathcal{D}}(Y[n], X) \to H(Y[n]) \text{ is an isomorphism for all } n\}$$

As $\text{Hom}_{\mathcal{D}}(-, X) \to H$ is a transformation between cohomological functors, the subcategory $\mathcal{D}'$ is a strictly full, saturated, triangulated subcategory of $\mathcal{D}$ (details omitted; see proof of Lemma [original label: `lemma-homological-functor-kernel`]). Moreover, as both H and $\text{Hom}_{\mathcal{D}}(-, X)$ transform direct sums into products, we see that direct sums of objects of $\mathcal{D}'$ are in $\mathcal{D}'$. Thus derived colimits of objects of $\mathcal{D}'$ are in $\mathcal{D}'$. Since $\{E_i\}$ is preserved under shifts, we see that E_i is an object of $\mathcal{D}'$ for all i. It follows from Lemma 5.3 that $\mathcal{D}' = \mathcal{D}$ and the proof is complete. □

Proposition 6.2. Let $\mathcal{D}$ be a triangulated category with direct sums which is compactly generated. Let $F : \mathcal{D} \to \mathcal{D}'$ be an exact functor of triangulated categories which transforms direct sums into direct sums. Then F has an exact right adjoint.

Proof. For an object Y of $\mathcal{D}'$ consider the contravariant functor

$$\mathcal{D} \to \text{Ab}, \quad W \mapsto \text{Hom}_{\mathcal{D}'}(F(W), Y)$$

This is a cohomological functor as F is exact and transforms direct sums into products as F transforms direct sums into direct sums. Thus by Lemma 6.1 we find an object X of $\mathcal{D}$ such that $\text{Hom}_{\mathcal{D}}(W, X) = \text{Hom}_{\mathcal{D}'}(F(W), Y)$. The existence of the adjoint follows from Categories, Lemma [original label: `categories-lemma-adjoint-exists`]. Exactness follows from Lemma [original label: `lemma-adjoint-is-exact`]. □

7 Brown representability, bis

In this section we explain a version of Brown representability for triangulated categories which have a suitable set of generators; for other versions, please see [2], [5], and [3].

Lemma 7.1. Let $\mathcal{D}$ be a triangulated category with direct sums. Suppose given a set $\mathcal{E}$ of objects of $\mathcal{D}$ such that

1. if X is a nonzero object of $\mathcal{D}$, then there exists an $E \in \mathcal{E}$ and a nonzero map $E \to X$, and
2. given objects X_n, $n \in \mathrm{N}$ of $\mathcal{D}$, $E \in \mathcal{E}$, and $\alpha : E \to \bigoplus X_n$, there exist $E_n \in \mathcal{E}$ and $\beta_n : E_n \to X_n$ and a morphism $\gamma : E \to \bigoplus E_n$ such that $\alpha = (\bigoplus \beta_n) \circ \gamma$.

Let $H : \mathcal{D} \to \mathrm{Ab}$ be a contravariant cohomological functor which transforms direct sums into products. Then H is representable.

Proof. This proof is very similar to the proof of Lemma 6.1. We may replace $\mathcal{E}$ by $\bigcup_{i \in \mathrm{Z}} \mathcal{E}[i]$ and assume that $\mathcal{E}$ is preserved by shifts. Consider pairs (E, a) where $E \in \mathcal{E}$ and $a \in H(E)$ and set

$$X_1 = \bigoplus\nolimits_{(E,a)} E$$

Since $H(X_1) = \prod_{(E,a)} H(E)$ we see that $(a)_{(E,a)}$ defines an element $a_1 \in H(X_1)$. Set $H_1 = \mathrm{Hom}_{\mathcal{D}}(-, X_1)$. By Yoneda's lemma (Categories, Lemma [original label: `categories-lemma-yoneda`]) the element a_1 defines a natural transformation $H_1 \to H$.

We are going to inductively construct X_n and transformations $a_n : H_n \to H$ where $H_n = \mathrm{Hom}_{\mathcal{D}}(-, X_n)$. Namely, we apply the procedure above to the functor $\mathrm{Ker}(H_n \to H)$ to get an object

$$K_{n+1} = \bigoplus\nolimits_{(E,k),\ k \in \mathrm{Ker}(H_n(E) \to H(E))} E$$

and a transformation $\mathrm{Hom}_{\mathcal{D}}(-, K_{n+1}) \to \mathrm{Ker}(H_n \to H)$. By Yoneda's lemma the composition $\mathrm{Hom}_{\mathcal{D}}(-, K_{n+1}) \to H_n$ gives a morphism $K_{n+1} \to X_n$. We choose a distinguished triangle

$$K_{n+1} \to X_n \to X_{n+1} \to K_{n+1}[1]$$

in $\mathcal{D}$. The element $a_n \in H(X_n)$ maps to zero in $H(K_{n+1})$ by construction. Since H is cohomological we can lift it to an element $a_{n+1} \in H(X_{n+1})$.

Set $X = \mathrm{hocolim} X_n$. Applying H to the defining distinguished triangle

$$\bigoplus X_n \to \bigoplus X_n \to X \to \bigoplus X_n[1]$$

we obtain an exact sequence

$$\prod H(X_n) \leftarrow \prod H(X_n) \leftarrow H(X)$$

Thus there exists an element $a \in H(X)$ mapping to (a_n) in $\prod H(X_n)$. Hence a natural transformation $\mathrm{Hom}_{\mathcal{D}}(-, X) \to H$ such that

$$\mathrm{Hom}_{\mathcal{D}}(-, X_1) \to \mathrm{Hom}_{\mathcal{D}}(-, X_2) \to \mathrm{Hom}_{\mathcal{D}}(-, X_3) \to \ldots \to \mathrm{Hom}_{\mathcal{D}}(-, X) \to H$$

commutes. We claim that $\mathrm{Hom}_{\mathcal{D}}(-, X) \to H(-)$ is an isomorphism.

Let $E \in \mathcal{E}$. Let us show that

$$\mathrm{Hom}_{\mathcal{D}}(E, \bigoplus X_n) \to \mathrm{Hom}_{\mathcal{D}}(E, \bigoplus X_n)$$

is injective. Namely, let $\alpha : E \to \bigoplus X_n$. Then by assumption (2) we obtain a factorization $\alpha = (\bigoplus \beta_n) \circ \gamma$. Since $E_n \to X_n \to X_{n+1}$ is zero by construction, we see that the composition $\bigoplus E_n \to \bigoplus X_n \to \bigoplus X_n$ is equal to $\bigoplus \beta_n$. Hence also the composition $E \to \bigoplus X_n \to \bigoplus X_n$ is equal to α. This proves the stated injectivity and hence also

$$\mathrm{Hom}_{\mathcal{D}}(E, \bigoplus X_n[1]) \to \mathrm{Hom}_{\mathcal{D}}(E, \bigoplus X_n[1])$$

is injective. It follows that we have an exact sequence

$$\mathrm{Hom}_{\mathcal{D}}(E, \bigoplus X_n) \to \mathrm{Hom}_{\mathcal{D}}(E, \bigoplus X_n) \to \mathrm{Hom}_{\mathcal{D}}(E, X) \to 0$$

for all $E \in \mathcal{E}$.

Let $E \in \mathcal{E}$ and let $f : E \to X$ be a morphism. By the previous paragraph, we may choose $\alpha : E \to \bigoplus X_n$ lifting f. Then by assumption (2) we obtain a factorization $\alpha = (\bigoplus \beta_n) \circ \gamma$. For each n there is a morphism $\delta_n : E_n \to X_1$ such that δ_n and β_n map to the same element of $H(E_n)$. Then the compositions

$$E_n \to X_n \to X_{n+1} \quad \text{and} \quad E_n \to X_1 \to X_{n+1}$$

are equal by construction of $X_n \to X_{n+1}$. It follows that

$$\bigoplus E_n \to \bigoplus X_n \to X \quad \text{and} \quad \bigoplus E_n \to \bigoplus X_1 \to X$$

are the same too. Observing that $\bigoplus X_1 \to X$ factors as $\bigoplus X_1 \to X_1 \to X$, we conclude that

$$\mathrm{Hom}_{\mathcal{D}}(E, X_1) \to \mathrm{Hom}_{\mathcal{D}}(E, X)$$

is surjective. Since by construction the map $\mathrm{Hom}_{\mathcal{D}}(E, X_1) \to H(E)$ is surjective and by construction the kernel of this map is annihilated by $\mathrm{Hom}_{\mathcal{D}}(E, X_1) \to \mathrm{Hom}_{\mathcal{D}}(E, X)$ we conclude that $\mathrm{Hom}_{\mathcal{D}}(E, X) \to H(E)$ is a bijection for all $E \in \mathcal{E}$.

To finish the proof, consider the subcategory

$$\mathcal{D}' = \{Y \in \mathrm{Ob}(\mathcal{D}) \mid \mathrm{Hom}_{\mathcal{D}}(Y[n], X) \to H(Y[n]) \text{ is an isomorphism for all } n\}$$

As $\mathrm{Hom}_{\mathcal{D}}(-, X) \to H$ is a transformation between cohomological functors, the subcategory $\mathcal{D}'$ is a strictly full, saturated, triangulated subcategory of $\mathcal{D}$ (details omitted; see proof of Lemma [original label: `lemma-homological-functor-kernel`]). Moreover, as both H and $\mathrm{Hom}_{\mathcal{D}}(-, X)$ transform direct sums into products, we see that direct sums of objects of $\mathcal{D}'$ are in $\mathcal{D}'$. Thus derived colimits of objects

of $\mathcal{D}'$ are in $\mathcal{D}'$. Since $\mathcal{E}$ is preserved by shifts, we conclude that $\mathcal{E} \subset \mathrm{Ob}(\mathcal{D}')$ by the result of the previous paragraph. To finish the proof we have to show that $\mathcal{D}' = \mathcal{D}$.

Let Y be an object of $\mathcal{D}$ and set $H(-) = \mathrm{Hom}_{\mathcal{D}}(-, Y)$. Then H is a cohomological functor which transforms direct sums into products. By the construction in the first part of the proof we obtain a morphism $\mathrm{colim}\, X_n = X \to Y$ such that $\mathrm{Hom}_{\mathcal{D}}(E, X) \to \mathrm{Hom}_{\mathcal{D}}(E, Y)$ is bijective for all $E \in \mathcal{E}$. Then assumption (1) tells us that $X \to Y$ is an isomorphism! On the other hand, by construction $X_1, X_2, \ldots$ are in $\mathcal{D}'$ and so is X. Thus $Y \in \mathcal{D}'$ and the proof is complete. □

Proposition 7.2. Let $\mathcal{D}$ be a triangulated category with direct sums. Assume there exists a set $\mathcal{E}$ of objects of $\mathcal{D}$ satisfying conditions (1) and (2) of Lemma 7.1. Let $F : \mathcal{D} \to \mathcal{D}'$ be an exact functor of triangulated categories which transforms direct sums into direct sums. Then F has an exact right adjoint.

Proof. For an object Y of $\mathcal{D}'$ consider the contravariant functor

$$\mathcal{D} \to \mathrm{Ab}, \quad W \mapsto \mathrm{Hom}_{\mathcal{D}'}(F(W), Y)$$

This is a cohomological functor as F is exact and transforms direct sums into products as F transforms direct sums into direct sums. Thus by Lemma 7.1 we find an object X of $\mathcal{D}$ such that $\mathrm{Hom}_{\mathcal{D}}(W, X) = \mathrm{Hom}_{\mathcal{D}'}(F(W), Y)$. The existence of the adjoint follows from Categories, Lemma [original label: `categories-lemma-adjoint-exists`]. Exactness follows from Lemma [original label: `lemma-adjoint-is-exact`]. □

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